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Jérôme David & Cassia Trojahn (UGA, LIG, Inria - MOEX Team)
Sophie Ebersold (IRIT, Toulouse)

Subject: Application for PhD Position – Federation of data and models to define digital twins (PC2)

Dear Jérôme David, Cassia Trojahn, and Sophie Ebersold,

I am highly motivated to apply for the PC2 PhD position focused on the "Federation of data and models to define digital twins," hosted by the MOEX team at INRIA/Grenoble and IRIT. As an engineer with a strong foundation in knowledge engineering, deep learning, and complex systems architecture, I am deeply interested in addressing the interoperability challenges that arise when coupling heterogeneous simulation models and disparate data sources.

In my engineering of the Industrial Gas Turbine Digital Twin, I faced the practical challenges of federation firsthand. The system required aligning complex thermodynamic equations (Navier-Stokes, Brayton cycle) with high-frequency telemetry streams and Edge-TCN anomaly models. Creating a coherent digital twin meant standardizing the semantics between the physical models and the data-driven models. To achieve this, I established rigorous modular boundaries and data pipelines that allowed each sub-system to communicate effectively. I am eager to formally scale this experience by researching ontology-based alignment and semantic mediation techniques. The goal of automatically aligning conceptual models and raw data schemas across different domains is a challenge I am well-prepared to tackle, ensuring that the Artemis platform can natively support federated architectures.

My academic background in Artificial Intelligence further positions me to explore machine learning-assisted semantic alignment, automating the process of mapping data points to model variables. I am excited to collaborate with the MOEX and IRIT teams to build robust federation mechanisms that will serve as the benchmark for future digital twin deployments. Please find attached my resume and academic transcripts. I welcome the opportunity to discuss my application further.

Sincerely,

Ismail Aissaoui